STAFF REVIEW · H446 220 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. They allow reconstruction of the actual shortest route.

2. No. Its shortest-path guarantee requires non-negative weights.

3. 7.

4. 6.

I Do — source worked steps

1. Predict the next state — From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.

2. Update the table — Through Q, R improves to g4,h2,f6; G gets g11.

3. Update the table — From R, G improves to g6,h0,f6.

4. Update the table — Select G: predecessor chain yields P→Q→R→G, cost 6.

We Do — guided CHECK

1. Update the table
From N, O improves to g4,h1,f5.
From O, G improves to g6,h0,f6.
Select G and reconstruct M→N→O→G, total cost 6.

You Do — source mark schemes

1. [4] f=10 [1]; g is known travel cost so far [1]; h is estimated remaining cost [1]; f is estimated total [1].

2. [6] From S, A has g2,h2,f4 [2]; B has g5,h1,f6 [2]. Expanding A improves B to g3,f4 [1]; route S→A→B→G total 5 [1].

3. [6] A heuristic is admissible when it never overestimates the true cheapest remaining cost [2]. From A the direct cheapest cost to G is 4 [2], so h(A) may be at most 4 [1] (assuming non-negative estimates); overestimating can lead to the optimal path being missed under early-goal termination [1].

Exit Ticket — source answer

1. False. It normally chooses the frontier vertex with smallest f(n)=g(n)+h(n).